INCH-POUND

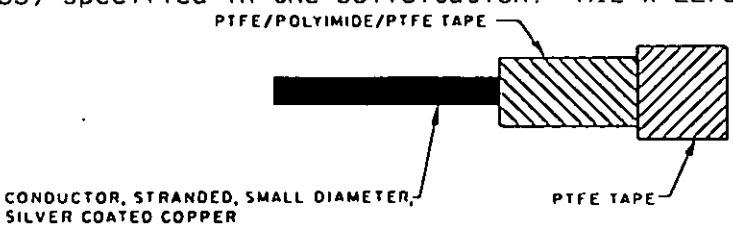
MIL-W-22759/91
9 September 1994

MILITARY SPECIFICATION SHEET

# WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE/POLYIMIDE INSULATED, LIGHT WEIGHT, SILVER COATED, COPPER CONDUCTOR, 200°C, 600 VOLTS

This specification sheet has been approved by all departments and agencies within the Department of Defense: however, it is not intended for use in Naval aircraft or Naval Air Systems applications.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-22759.




PTFE - POLYTETRAFLUOROETHYLENE

FIGURE 1. General configuration.

TABLE I. Construction details.

| Part No. 1 | Wire Size | Conductor | | | Finished Wire | | | |
|---|---|---|---|---|---|---|---|---|
| | | Stranding (Number of Strands X AWG Gauge of Strands) | Diameter (in.) | | Resistance at 20°C (68°F) (ohms/1000 ft)(Max) | Diameter (in.) | | Weight (lb/1000 ft) (Max) |
| | | | Min | Max | | Min | Max | |
| M22759/ -26-* | 26 | 19 x 38 | 0.018 | 0.020 | 38.4 | 0.030 | 0.034 | 1.3 |
| M22759/ -24-* | 24 | 19 x 36 | 0.023 | 0.024 | 24.3 | 0.034 | 0.038 | 1.9 |

AMSC N/A FSC 6145

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TABLE I. Construction details - Continued.

| Part No. 1/ | Wire Size | Conductor | | | Finished Wire | | | |
|---|---|---|---|---|---|---|---|---|
| | | Stranding (Number of Strands X AWG Gauge of Strands) | Diameter (in.) | | Resistance at 20°C (68°F) (ohms/1000 ft) (Max) | Diameter (in.) | | Weight (lb/1000 ft) (Max) |
| | | | Min | Max | | Min | Max | |
| M22759/ -22-* | 22 | 19 x 34 | 0.029 | 0.030 | 15.1 | 0.040 | 0.043 | 2.8 |
| M22759/ -20-* | 20 | 19 x 32 | 0.037 | 0.038 | 9.19 | 0.048 | 0.051 | 4.3 |
| M22759/ -18-* | 18 | 19 X 30 | 0.046 | 0.048 | 5.79 | 0.056 | 0.060 | 6.5 |
| M22759/ -16-* | 16 | 19 X 29 | 0.052 | 0.054 | 4.52 | 0.065 | 0.069 | 8.4 |
| M22759/ -14-* | 14 | 19 X 27 | 0.065 | 0.068 | 2.88 | 0.078 | 0.082 | 12.8 |
| M22759/ -12-* | 12 | 37 X 28 | 0.084 | 0.087 | 1.90 | 0.097 | 0.101 | 19.6 |
| M22759/ -10-* | 10 | 37 X 26 | 0.106 | 0.110 | 1.19 | 0.120 | 0.124 | 30.8 |

1/ Part number: The asterisks in the part number column of Table I shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white with orange stripe - M22759/ -20-93.

TABLE II. Wire insulation materials. 1/

| Tape Code | Thickness (Nominal) | Material |
|---|---|---|
| 1 | 0.0015 | 0.00065(PTFE)/0.00075(Polyimide)/0.0001(PTFE) or 0.00038(PTFE)/0.00075(Polyimide)/0.00038(PTFE) |
| 2 | 0.0015 | PTFE (Unsintered or presintered bondable) |
| 3 | 0.002 | PTFE (Unsintered or presintered bondable) |
| 4 | 0.0025 | PTFE (Unsintered or presintered bondable) |

1/ Physical properties of PTFE tapes (skived and unsintered) shall be in accordance with MIL-W-22759 requirements.



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TABLE III. Physical properties of PTFE/Polyimide/PTFE tape.

| | |
|---|---|
| Tensile Strength | 11,000 lb/in$^2$ (average minimum) |
| Elongation | 40 percent (average minimum) |
| Dielectric Strength | 3,000 volts/mil (average minimum) |

TABLE IV. Tape overlap requirements. 1/

| Wire Size | Wrap 1 | | Wrap 2 | | Nominal Wall Thickness |
|---|---|---|---|---|---|
| | Tape Code | Percent Overlap (minimum) | Tape Code | Percent Overlap (minimum) | |
| 26 | 1 | 50 | 2 | 50 | 0.0058 |
| 24 | 1 | 50 | 2 | 50 | 0.0058 |
| 22 | 1 | 50 | 2 | 50 | 0.0058 |
| 20 | 1 | 50 | 2 | 50 | 0.0058 |
| 18 | 1 | 50 | 2 | 50 | 0.0058 |
| 16 | 1 | 50 | 3 | 50 | 0.007 |
| 14 | 1 | 50 | 3 | 50 | 0.007 |
| 12 | 1 | 50 | 4 | 50 | 0.0075 |
| 10 | 1 | 50 | 4 | 50 | 0.0075 |

1/ Wrap 1 is innermost tape which is in contact with the conductor with the 0.00065 inch PTFE side of the tape against the conductor (when applicable).



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TABLE V. Fluid table.

| Test fluid | Test temperature (°C(°F)) | Immersion time (hrs) |
|---|---|---|
| A. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, undiluted | 48-50 (118-122) | 20 |
| B. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, diluted 60/40 (fluid/water) ratio | 48-50 (118-122) | 20 |
| C. MIL-C-43616, Cleaning Compound, Aircraft Surface | 48-50 (118-122) | 20 |
| D. ASTM D1153, Methyl Isobutyl Keyton (For use in Organic Coatings) | 20-25 (68-77) | 168 |
| E. SAE-AS-1241, Fire Resistant Hydraulic Fluid for Aircraft | 48-50 (118-122) | 20 |
| F. MIL-L-7808, Lubricating Oil, Aircraft Turbine Engine, Synthetic Base | 118-121 (244-250) | 30 |
| G. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline Waterbase, undiluted | 63-68 (145-154) | 20 |
| H. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline Waterbase, diluted 25/75 (fluid/water) ratio | 63-68 (145-154) | 20 |
| I. TT-S-735, Standard Test Fluids: Hydrocarbon, Type I | 20-25 (68-77) | 168 |
| J. TT-S-735, Standard Test Fluids: Hydrocarbon, Type II | 20-25 (68-77) | 168 |
| K. TT-S-735, Standard Test Fluids: Hydrocarbon, Type III | 20-25 (68-77) | 168 |
| L. TT-S-735, Standard Test Fluids: Hydrocarbon, Type VII | 20-25 (68-77) | 168 |



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TABLE V. Fluid table - Continued.

| Test fluid | Test temperature (°C(°F)) | Immersion time (hrs) |
|---|---|---|
| M. Dielectric-coolant fluid, synthetic silicate ester base, Monsanto coolanol 25 or approved equivalent | 20-25 (68-77) | 168 |
| N. MIL-G-3056, Gasoline, Automotive, Combat | 20-25 (68-77) | 168 |

RATINGS:

*Temperature rating:* 200°C (392°F) maximum continuous conductor temperature.
Voltage rating: 600 Volts (rms) at sea level.

ADDITIONAL REQUIREMENTS:

Wet arc propagation resistance (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 3006. Measure the damage of the bundle along the axis. The wire is acceptable if the following criteria are met:

1. A minimum of 64 wires pass the dielectric test.
2. Three wires or less fail the dielectric test in any one bundle.
3. Actual damage to the wire is not more than 3 inches in any test bundle.

Dry arc propagation resistance (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 3007. Measure the damage of the bundle along the axis. The wire is acceptable if the following criteria are met:

1. A minimum of 64 wires pass the dielectric test.
2. Three wires or less fail the dielectric test in any test bundle.
3. Actual damage to the wire is not more than 3 inches in any test bundle.

Blocking: 200°C ± 2°C (392°F ± 3.6°F).

Color: In accordance with MIL-STD-104, class 1; white preferred. Conformity of color to the limits of MIL-STD-104 shall not be required after oven exposure.

Color striping or banding durability: 125 cycles (250 strokes), 250 grams weight.



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Conductor strand adhesion: Required.

Continuous lengths: Schedule B.

Dynamic cut-through (Test required for initial qualification only): Test size 20 only in accordance with ASTM D3032 section 22. Blade shall be the standard blade except the cutting edge radius shall be .005 ± .001 inch. Minimum dynamic cut-through of 14 lbs at 23 ± 2°C (73°F ± 3.6°F) and 5 lbs at 200 ± 2°C (392°F ± 3.6°F).

Flammability: Test in accordance with MIL-STD-2223 method 1006 procedure A. Requirements:

| | |
|---|---|
| Duration of after-flame | 3 seconds (max) |
| Flame travel | 3.0 inches (max) |
| No flaming of tissue | |

*High-frequency spark test: (When used in lieu of Impulse Dielectric Test)* Test in accordance with MIL-STD-2223 method 3008. 5.7 kilovolts (rms). Test 100 percent of the wire.

Humidity resistance: After humidity exposure, wire shall meet the requirements for initial insulation resistance.

Identification of product: Not required for size 26. Color code designator not required.

Identification durability: 125 cycles (250 strokes), 250 grams weight.

Immersion (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 1001 including additional fluids listed in Table V of this specification. Use mandrels and weights listed in Table VI for bend testing. Dielectric test, 2500 volts (rms), 60 Hz. For the turbine fuel immersion test of MIL-STD-2223, either Type JP-4 or MIL-T-83133 Type JP-8 (NATO Type F-34) may be used.

Impulse dielectric test: 8.0 kilovolts (peak). Test 100 percent of the wire.

Insulation resistance: 5000 megohms for 1000 feet (min).

Life cycle: 500 hours at 230°C ± 2°C (446°F ± 3.6°F). Dielectric test, 2500 volts (rms), 60 Hz. Use mandrels coated with polytetrafluoroethylene such that the diameter of the mandrels, after coating, still conform to the required test mandrel diameters in Table VI. After oven exposure, layers shall not separate and/or tapes shall not lift along the insulation or at the ends.

Low temperature (cold bend): Use mandrels and weights specified in Table VI. Chamber temperature, -65°C ± 2°C (-85°F ± 3.6°F). Dielectric test, 2500 volts (rms), 60 Hz.



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Shrinkage: 0.091 inch (max) at 230°C ± 2°C (446°F ± 3.6°F).

Smoke: 200°C ± 5°C (392°F ± 9°F); no visible smoke.

Solderability: Required.

Strippability: (Group II quality conformance test). Test size 26-14 only in accordance with ASTM D3032 section 27. The length of the insulation slugs shall be 0.25 inches. The maximum strip force shall be 4 lbs for sizes 20-26 and 6 lbs for sizes 18-14. No evidence of insulation left on the conductor when viewed with the naked eye.

Thermal Index: 200°C (392°F) minimum for 10,000 hours (Test required for initial qualification only). Test size 20 only in accordance with ASTM D 3032, section 14.

Thermal shock resistance: Oven temperature, 200°C ± 2°C (392°F ± 3.6°F). Maximum change in measurement, 0.091 inches. No cracking.

Wrap (mandrel wrap): No cracking, no dielectric breakdown. Use mandrels specified in Table VI. No wrinkles shall be visible in the insulation at 3X magnification. Dielectric test, 2500 volts (rms), 60 Hz.

TABLE VI. Test mandrel and test load requirements.

| Wire Size (AWG) | Test Mandrel Diameter 1/ (in.) | | | Test Load 1/ (lb) | |
|---|---|---|---|---|---|
| | Cold Bend | Life Cycle/ Bend test | Wrap | Cold Bend | Life Cycle/ Bend test |
| 26 | 1.00 | 0.375 | 0.125 | 3.00 | 0.50 |
| 24 | 1.00 | 0.500 | 0.125 | 3.00 | 0.75 |
| 22 | 1.00 | 0.500 | 0.125 | 4.00 | 1.00 |
| 20 | 1.00 | 0.500 | 0.125 | 4.00 | 1.50 |
| 18 | 1.50 | 0.750 | 0.250 | 5.00 | 2.00 |
| 16 | 1.50 | 1.00 | 0.250 | 5.00 | 2.00 |
| 14 | 2.00 | 1.00 | 0.375 | 5.00 | 3.00 |
| 12 | 2.00 | 1.50 | 0.375 | 5.00 | 3.00 |
| 10 | 3.00 | 2.00 | 0.375 | 6.00 | 3.00 |

1/ Tolerance shall be ± 3 percent of the given values.



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Qualification of wire:

For qualification, a source is required to submit data on quality conformance tests and any finished wire tests as required by the qualification authorization letter. All other testing will be performed by the qualifying activity at the source's expense.

Due to the extended time period over which the thermal index test is performed, a source may become qualified under this specification sheet while this test is still in progress.

Custodian:
Army - CR
Navy - AS
Air Force - 85
NASA - NA

Review Activities:
Army - AV, MI
Air Force - 11, 19, 70, 99
DLA - ES, IS

Preparing Activity
Navy - AS

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